VIETNAM GENERAL CONFEDERATION OF LABOR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

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MIDTERM REPORT

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

HO CHI MINH CITY, 2025

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INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Advised by

Ths. Nguyen Thanh An

HO CHI MINH CITY, 2025

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Ho Chi Minh city, 4th January 2025.

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DECLARATION OF AUTHORSHIP

We hereby declare that this is our own project and is guided by Ths. Nguyen Thanh An; The content research and results contained herein are central and have not been published in any form before. The data in the tables for analysis, comments and evaluation are collected by the main author from different sources, which are clearly stated in the reference section.

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- Assigned task: do task 2, give feedback on task 1 and 3, make presentation.pdf
- Complete percentage: 90%

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- Complete percentage: 90%

TABLE OF CONTENT

TASK COMPLETION TABLE iii

CHAPTER 1. FORMULATION PROBLEMS.....1

1.1 Task 1:.....1

1.2 Task 2:.....1

1.3 Task 3:.....2

CHAPTER 2. SOLUTIONS.....3

2.1 Task 1:.....3

2.2 Task 2 7

2.3 Task 3:.....8

REFERENCES.....10

TASK COMPLETION TABLE

TASK	PERCENT COMPLETE
1.	90%
2.	90%
3.	90%

CHAPTER 1. FORMULATION PROBLEMS

1.1 Task 1:

- State : the location of each of the eight tiles and the blank
- Initial State : can be arbitrary permutation of 8 tiles and the blank cell.
- Actions : movements of the blank space
  - Left, Right, Up, or Down.
  - Different subsets of these are possible depending on where the blank is.
  - After a move :
    - If cell 1 and cell 3 are adjacent to each other, horizontally or vertically, they are automatically swapped.
    - Similar for cell 2 and cell 4.
- Transition model : return a resulting state given a state and an action, measure the average path cost of the solution.
- Goat test : there are four goal states as below.

1	2	3
4	5	6
7	8	

8	7	6
5	4	3
2	1	

	1	2
3	4	5
6	7	8

	8	7
6	5	4
3	2	1

- Path cost : each step costs 1.

1.2 Task 2:

- State : the maze structure is as below :
  - % → obstacles/walls
  - P → initial location of the pacman
  - . → food points, there could be multiple points
  - O → magical pies, pacman can go through walls for the 5 steps after eating a pie. If it's stuck on the wall after 5 steps → "Game over".
  - spaces → blank cells.
  - There are no ghosts in the maze.
  - If the pacman reaches to a corner, then it automatically teleports to the opposite corner of the maze.
- Initial State : a path where pacman stands at any space on the path

  Example:

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%         .          P             %
% %%%%%%%%%%%%%%%%%%%%%%%% %%%%%%%% %
% %%   %   %       %%%%%%%   %%0    %
% %% % % % % %%%% %%%%%%%%% %% %%%%%
% %% % % % % .             %% %%     %
% %% % % % % % %%%%  %%%  . %%%%%% %
% %  % % %   %    %% %%%%%%%%       %
% %% %0% %%%%%%%% %%    .   %% %%%%%
% %% %   %%        %%%%%%%%% %%     %
%    %%%%%% %%%%%%%   .  %% %%%%%% %
%%%%%%       %        %%%% %% %0     %
%  0   %%%%%% %%%%% %    %% %% %%%%%
% %%%%%% .    %        %%%%% %%     %
%         %%%%%% %%%%%%%%%%% %%  %% %
%%%%%%%%%%                   %%%%%% %
%.          %%%%%%%%%%%%%%%%         %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

- Actions : North, East, West, South, Stop
- Transition model : list of actions (North, East, West, South, Stop); total cost
- Goal test : the pacman collects all food points in the maze.
- Path cost : each step costs 1.

1.3 Task 3:

- States: any arrangement of 0 to16 queens on the board
- Initial state: no queens on the board
- Actions: add a queen to any empty square
- Transition model: returns the board with a queen added to the specified square
- Goal test: 16 queens are on the board, none attacked :
  - $256 \cdot 255 \cdots 241 \approx 2.108756021 \times 10^{38}$ possible sequences to investigate
- Path cost : trivial because only the final state counts.

CHAPTER 2. SOLUTIONS.

2.1 Task 1:

TWO HEURISTIC FUNCTIONS:

- **Manhattan distance:**

- Idea:

The Manhattan Distance heuristic computes the total distance by summing the horizontal and vertical distances between the current position of each tile (excluding the blank) and its corresponding goal position.

For each tile, the formula is:

distance $= \left|x_{hiện\ tại} - x_{mục\ tiêu}\right| + \left|y_{hiện\ tại} - y_{mục\ tiêu}\right|$

- Example:

Current State: 1 2 3
4 8 5
7 _ 6

Goal State: 1 2 3
4 5 6
7 8 _

Tile 8:

Current position: (2,1) → Goal position: (3,2)

Distance = |2-3| + |1-2| = 1 + 1 = 2

Tile 5:

Current position: (2,3) → Goal position: (2,2)

Distance = |2-2| + |3-2| = 0 + 1 = 1

Tile 6:

Current position: (3,3) → Goal position: (2,3)

Distance = |3-2| + |3-3| = 1 + 0 = 1

Total Manhattan Distance (heuristic) = 2 + 1 + 1 = 4

- Admissibility:

The total Manhattan Distance for all tiles represents the minimum number of moves required if each move brings a tile closer to its goal position. Since every actual move in the puzzle must cover at least one unit of this distance, the Manhattan Distance never overestimates the true minimum cost to reach the goal.

Conclusion: Admissible.

- Consistency:

When performing a move, the Manhattan Distance for any tile changes by at most 1 (if the move brings it closer to its target). This means that the difference between h(n) and h(n′) will not exceed the cost of the move.

Conclusion: Consistent.

- Pseudocode:

```
function MANHATTAN-DISTANCE(state, goal) return heuristic
    distance ← 0
    for each tile in state DO:
      if tile ≠ empty THEN:
        (x_current, y_current) ← current position of tile in state
        (x_goal, y_goal) ← correct position of tile in goal
        distance ← distance + (ABS(x_current - x_goal) +
                               ABS(y_current - y_goal))
    return distance
```

- **Misplaced titles:**

- Idea:

Count the tiles that are not in the correct position as defined by the goal state. If a tile is not in its correct position and is not the blank, it is considered "misplaced."

- ví dụ:

Current State: 1 2 3 **Goal State**: 1 2 3

4 8 5 4 5 6

7 _ 6 7 8 _

The misplaced tiles are 8, 5 và 6.

Misplaced tiles (heuristic) = 3.

- **Admissibility:**

Every misplaced tile requires at least one move to be corrected. Therefore, the total count of misplaced tiles will always be less than or equal to the actual cost needed to reach the goal state.

Conclusion: Admissible..

- **Consistency:**

With each move, the change in the number of misplaced tiles cannot exceed 1 (given that the move cost is 1). This ensures that the difference between the heuristic values of successive states does not exceed the cost of the move.

Conclusion: Consistent.

- **Pseudocode**:

```
function MISPLACED-TILES(state, goal) return Heuristic value
    count ← 0
      for each tile in state DO:
            IF tile ≠ empty AND tile is not in its correct position
            according to goal THEN:
                count ← count + 1
        return count
```

- **Pros and cons:**

Heuristic Fuctions	Advantages	Disadvantages
Manhattan Distance	- Provides a closer lower bound for the actual cost. - Reflects both horizontal and vertical moves.	- Slightly more computationally expensive than simple counts.
Misplaced Tiles	- Simple and fast to compute. - Easy to implement.	- May provide a less accurate cost estimate. - Slightly more computationally expensive than simple counts.

PSEUDOCODE A* SEARCH:

```
function A_STAR(problem, heuristic) return solution or failure
    node ← initial state
    frontier ← a heap with key is g(n) + h(n)
    set ← empty set
    WHILE frontier
        current ← heap pop
        IF current is goal state
            return solution
        set add current
        FOR successor in current successors:
            frontier add successor
    return failure
```

2.2 Task 2

DIVIDING THE PROBLEM INTO TWO SUB-CLASSES

- **Map Reader Class:**

  Responsible for reading and parsing the map. This class extracts the layout and details of the environment.

- **Problem Information Class:**

  Stores all relevant information for the problem, including:

  - **Pacman's Position**: The current location of Pacman.
  - **Food Items' Positions**: Locations of the food that Pacman needs to eat.
  - **Magic Pies' Positions**: Locations of special items (magic pies).
  - **Possible Actions**: The set of actions Pacman can take (e.g., up, down, left, right).
  - **isTeleport Condition**: return teleport position if pacman is in corners

PSEUDOCODE A* SEARCH:

```
function A_STAR(problem, heuristic) return solution or failure
    node ← initial state
    frontier ← a heap with key is g(n) + h(n)
    set ← empty set
    WHILE frontier
        current ← heap pop
        IF current is goal state
            return solution
        set add current
        FOR successor in current successors:
            frontier add successor
    return failure
```

VISUALIZATION WITH PYGAME

- **Purpose:**
  Utilize the Pygame library to visualize and animate Pacman's movement as he navigates through the map using the A* search algorithm.
- **Implementation Overview:**
  - **Map Display:** Render the game map including walls, food items, and magic pies.
  - **Animation:** Animate Pacman's movement step-by-step along the computed path.
  - **Real-time Updates:** Update the display to reflect Pacman's current position and path progression.

2.3 Task 3:

GENETIC ALGORITHM APPROACH

Data Structures

- **Queen:** Represents a queen with row and column attributes.
- **Board:** Contains a list of queens. Each queen's column is initialized randomly for its corresponding row.

Fitness Function

- **Calculation:**
  - Iterate through each pair of queens and count the number of attacking pairs (same column or diagonal).
  - Fitness = Maximum possible pairs − Number of attacking pairs.
  - A perfect solution is achieved when fitness reaches the maximum (i.e., no attacks).

Crossover (Two-Point Crossover)

- **Mechanism:**
  - Randomly select two crossover points (c1 and c2).

- Copy segments: the first segment and the third segment from the first parent, and the middle segment from the second parent.

Mutation

- With a defined probability, randomly alter the column position of some queens to maintain genetic diversity.

Genetic Process

1. **Initialization:** Generate an initial population of boards with queens placed in random column positions.
2. **Selection:** Use tournament selection (with a tournament size of 5) to choose parents.
3. **Crossover & Mutation:** Create offspring through crossover and apply mutation according to a preset mutation rate.
4. **Iteration:** Update the population across several generations until an optimal solution is found or a maximum number of generations is reached.

Pros and cons of genetic algorithm

Advantages	**Disadvantages**
- Easily adaptable to various optimization problems.	- Performance is sensitive to parameters (population size, mutation rate, number of generations).
- Mutation helps avoid local optima.	- Does not guarantee a globally optimal solution.
	- Requires multiple evaluations of the fitness function.

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